

Task 1: Understanding the Data

1. Introduction

The objective of this task is to perform an initial exploration of the dataset to understand its structure, size, and data quality. This includes identifying data types, missing values, duplicate records, and basic statistical insights to prepare the dataset for further analysis.

2. Dataset Overview

- **Dataset Name:** train_dataset.csv
- **Total Records (Rows):** 23
- **Total Features (Columns):** 5

3. Column Description

Column Name Description Type

Train Name	Categorical
Start Station	Categorical
End Station	Categorical
Distance	Numerical
Stops	Numerical

4. Data Type Information

- **Categorical Columns:** Train Name, Start Station, End Station
- **Numerical Columns:** Distance, Stops

This combination allows analysis of both route information and travel metrics.

5. Missing Values Analysis

Column Name Missing Values

Train Name 0

Start Station 1

End Station 0

Distance 1

Stops 1

Observation:

The dataset contains a small number of missing values across certain columns. These missing values should be handled during the data cleaning stage using appropriate imputation or removal techniques.

6. Duplicate Records Analysis

- **Total Duplicate Rows Found: 1**

Observation:

One duplicate record was identified in the dataset, which should be removed to maintain data integrity and avoid bias in analysis.

7. Basic Insights

- The dataset contains train route information including origin, destination, distance, and number of stops.
- Most records appear consistent and structured.
- Minor data quality issues such as missing values and duplication were detected.

8. Conclusion

The dataset has been successfully explored and its structure understood. It consists of 23 records and 5 attributes, with a mix of categorical and numerical data. While the dataset is largely clean, minor preprocessing steps such as handling missing values and removing duplicates are recommended before further analysis or modeling.